



Fracture of Ag sintering bonding interface on active metal brazing substrates for power electronics

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ABSTRACT

This study investigated the effects of Cu and Ag surface treatments on active metal brazing (AMB) substrates from two manufacturers (A and B) with respect to the sintered bonding characteristics of Ag paste and film. Surface roughness analysis showed that the Ag surface from manufacturer A had higher roughness than the Cu surface, whereas both Cu and Ag surfaces from manufacturer B exhibited comparable roughness values. Ag-paste bonding was highly sensitive to the surface roughness, leading to interfacial defects and void formation. By contrast, Ag-film bonding minimized the influence of surface roughness through a pressure sintering process, forming a uniform and dense interfacial structure. X-ray analysis revealed voids in the Ag-paste bond, whereas voids in the Ag-film bond were smaller than the detectable threshold. Scanning electron microscopy analysis after shear testing revealed that paste bonding resulted in substrate-side fractures, whereas film bonding resulted in chip-side fractures, indicating its higher bond strength. This study demonstrated that Ag film combined with pressure sintering can provide highly reliable bonding quality for high-power electronic module design while also offering key insights into the optimization of bonding processes.

1. Introduction

Sintered bonding materials with high thermal conductivity and mechanical stability are essential for effectively packaging high-power semiconductor devices. Recently, Ag-based sintering materials have gained significant attention in power semiconductor packaging owing to their high reliability, even at relatively low processing temperatures [1]. Active metal brazing (AMB) substrates are also widely used in high-power electric-vehicle modules and electronic devices, as they provide excellent thermal conductivity and mechanical stability [2]. The surface treatment of these substrates plays a crucial role in their interactions with the bonding materials, directly affecting the bonding quality [3]. Specifically, Ag paste and Ag film exhibit different bonding outcomes depending on sintering conditions and substrate surface properties [4]. Previous studies have reported that Ag paste exhibits high interfacial fluidity, making the interfacial quality highly sensitive to surface roughness [5]. In practice, Ag paste is applied not only in pressureless sintering but also in pressure-assisted sintering,

low-temperature bonding, and various power electronics packaging applications. In this study, we focused on pressureless sintering as a representative application to systematically compare it with pressure-assisted Ag-film bonding under controlled conditions. By contrast, Ag film, which is supplied in a solid form, has the potential to form a densely packed and uniform bonding interface through a pressure sintering process [6]. However, systematic studies on how these two bonding materials interact with substrates from different manufacturers with varied surface treatments, as well as how they affect bonding quality, remain limited [7]. Therefore, a comparison across supplier-dependent AMB surfaces and Cu/Ag finishes is needed. Such a comparison can establish interface-informed guidance for selecting bonding form and process conditions in high-power modules.

Particularly, interfacial nonuniformity is a critical factor influencing bonding quality, resulting in localized voids, gaps, and irregular distribution of bonding materials [8]. Such nonuniformity can cause stress concentration, thereby reducing bond strength; its effects vary significantly depending on the bonding material properties and substrate

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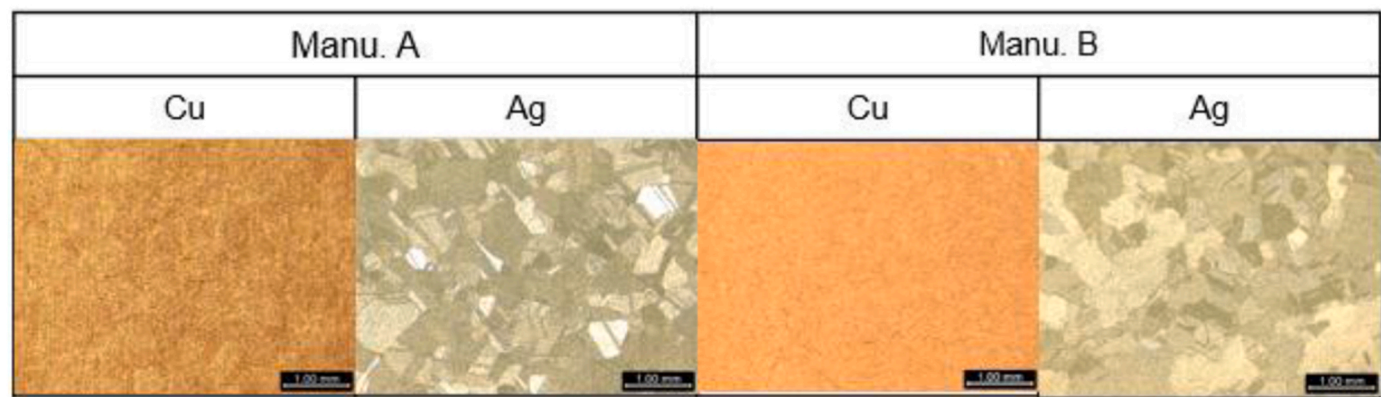


Fig. 1. Optical images of substrate surfaces (Manu. Stands for manufacturer).



Fig. 2. Surface profilometer (VK-9710, KEYENCE).

surface conditions [9]. From a surface and interface science perspective, real contact formation and void evolution during sintering are governed

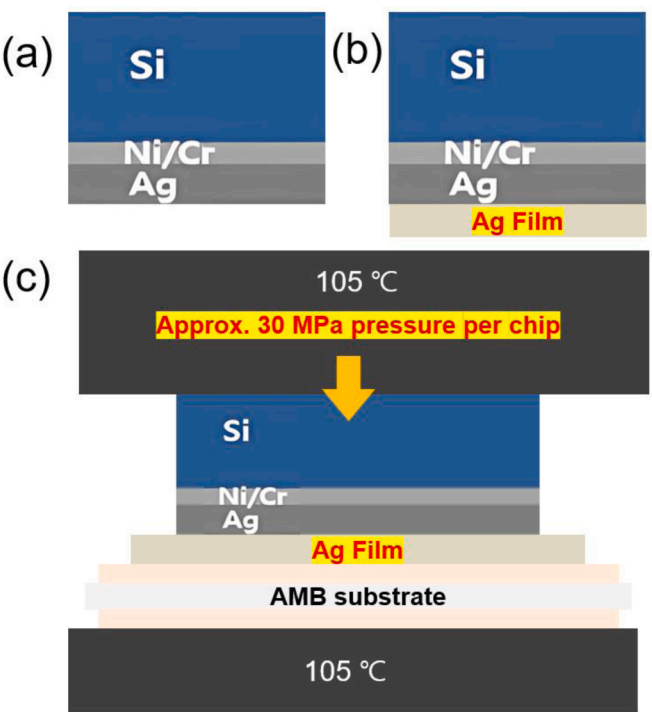


Fig. 3. Schematics of (a) the initial Si chip, (b) the Si chip after Ag film transfer, and (c) the Ag film transfer process onto the chip.

by the combined effects of surface topography and surface chemistry, as well as diffusion-driven microstructural evolution. Therefore, correlating interfacial defects with fracture location provides mechanistic insight into how interfacial nonuniformity influences joint reliability. In Ag-paste bonding, the high fluidity of the material leads to pronounced interactions with surface roughness, resulting in localized defects. By contrast, Ag-film bonding minimizes interfacial nonuniformity through pressure sintering, yielding a dense and uniform bonding interface [10].

This study systematically evaluated the bonding quality of Cu- and Ag-based AMB substrates from two manufacturers, designated as A and B. After separately sintering the Ag paste and Ag film, the characteristics of the bonding interfaces were analyzed using scanning electron microscopy (SEM), X-ray analysis, and shear strength testing. In addition, electron backscatter diffraction (EBSD) analysis was conducted to examine the microstructure of the bonding interface and to investigate the effects of grain-boundary structures and grain-size variations on bonding quality. Key observations indicate that Ag-paste bonding is highly sensitive to substrate surface states, resulting in interfacial

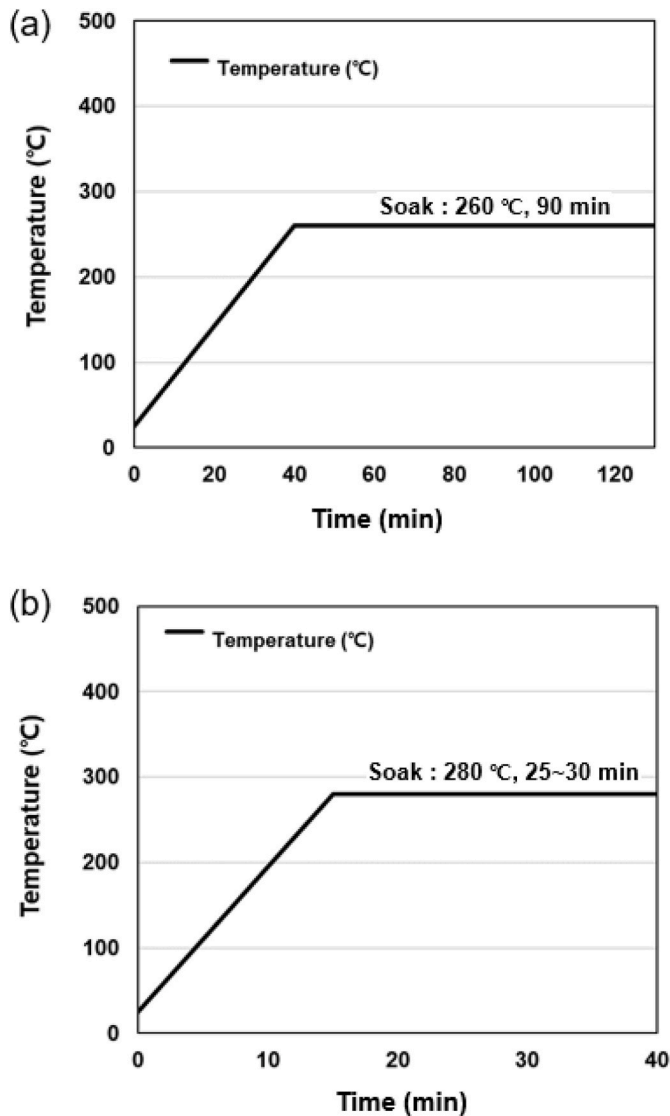


Fig. 4. Sintering conditions for (a) Ag paste and (b) Ag film.

defects and X-ray detectable voids, whereas pressure-assisted Ag-film bonding forms a dense, uniform interface with voids smaller than the X-ray detection threshold. Shear testing further indicates a fracture location shift from the substrate side (paste) to the chip side (film), consistent with higher interfacial integrity. The EBSD-based crystallographic analysis is used to relate interfacial microstructure (grain structure and misorientation) to bonding performance, providing guidance for reliable die-attach design in high-power electronic modules.

2. Experimental method

2.1. Preparation of materials and specimens

A total of four substrates were used in this study. The AMB substrates from two manufacturers, A and B, were selected, with each substrate categorized based on Cu and Ag surface treatments. Fig. 1 presents optical images of the substrates used in the experiment, categorized by manufacturer (A or B) and surface treatment (Cu or Ag). The Si chip used in this experiment has dimensions of 3 mm × 3 mm and features a layered structure: a silicon layer approximately 180 μm thick, followed by a Ni/Cr layer (1 μm thick), and finally an Ag layer (2 μm thick) at the bottom. During bonding, either Ag paste or Ag film was placed between the Ag-coated bottom surface of the Si chip and the substrate to form the

bonding interface. The Ag paste used (Kyocera, CT2700R7S) consists of elongated flake-shaped Ag particles sized approximately 7–8 μm mixed with smaller spherical particles sized approximately 2–3 μm, with an average particle size (D50) of approximately 5.50 μm. The Ag film used is a nanoparticle-based commercial Ag film with a thickness of approximately 65 μm. The substrates used in this experiment were Cu and Ag AMB substrates manufactured by companies A and B. The upper Cu layer thickness was 300 μm, the lower Cu layer thickness was 250 μm, and the ceramic layer in between was silicon nitride (Si₃N₄) with a thickness of approximately 320 μm. The thickness of the Ag coating on the substrates was approximately 0.6–0.8 μm. The paste is provided in a cream-like state, making it suitable for filling surface irregularities, whereas the film, supplied in solid form, enables the formation of a denser and more uniform bonding interface.

The surface roughness of the substrate is a crucial factor affecting the bonding quality. The average surface roughness (Ra) of the AMB substrate was measured in 3D using a laser confocal microscope (VK-9710, KEYENCE). Fig. 2 shows the instrument used for the surface-roughness measurement.

Ag sinter paste was printed onto the substrate using a metal mask with an 80% open area and a thickness of 80 μm, followed by the placement of the Si chip on the paste. By contrast, the Ag film was first bonded to the chip side rather than the substrate. Fig. 3 illustrates the Si chip condition before and after the Ag-film transfer, as well as the transfer conditions. Fig. 3(a) shows the Si chip before the Ag-film transfer; Fig. 3(b) shows the Si chip with the film uniformly transferred; and Fig. 3(c) presents the Ag-film transfer process conditions. The film was attached to the chip surface under optimized pressure and temperature conditions to ensure uniform bonding.

2.2. Sintering conditions and bonding characterization

The sintering process was conducted separately for the Ag paste and Ag film under distinct conditions. Fig. 4(a) shows the Ag paste sintered at 260 °C under a nonpressure N₂ atmosphere, whereas Fig. 4(b) shows the Ag film sintered at 280 °C under a uniaxial pressure of 10 MPa in an N₂ atmosphere. The equipment setup and schematics of the sintering processes are shown in Fig. 5, where Fig. 5(a) and (b) represent the paste and film sintering procedures, respectively.

Shear strength is a key indicator for evaluating the bonding quality. Fig. 6 shows the equipment and procedure used for the shear test. Fig. 6 (a) presents a schematic of the shear test, and Fig. 6(b) shows an image of the actual shear test equipment. Shear strength was measured using eight samples. The void ratio (%) was analyzed using an X-ray inspection system (X-TEK SYSTEMS LTD, Revolution), which is capable of detecting voids with a minimum diameter of approximately 30 μm. The void ratio was calculated by dividing the total void area by the total chip area (3 mm × 3 mm). Additionally, Fig. 7 illustrates the method used to measure the high bond ratio (HBR), defined as the proportion of well-bonded (void-free) areas relative to the total length of the bonding interface.

In this study, EBSD was employed to quantitatively analyze the grain misorientation at the bonding interface. The EBSD measurements were conducted using a velocity super system (EDAX), and the data were analyzed using TSL OIM Version 8 software. The EBSD measurements were used to assess the misorientation angle distribution between grains to evaluate interface uniformity and grain growth characteristics. The bonding interfaces were visualized in box plots to compare the differences between Cu and Ag substrates.

3. Results and discussion

3.1. Bonding Performance of Ag Paste and Ag Film

As shown in Fig. 8, the Ag substrate from manufacturer A (roughness of approximately 1.95 μm) is rougher than that of its Cu counterpart

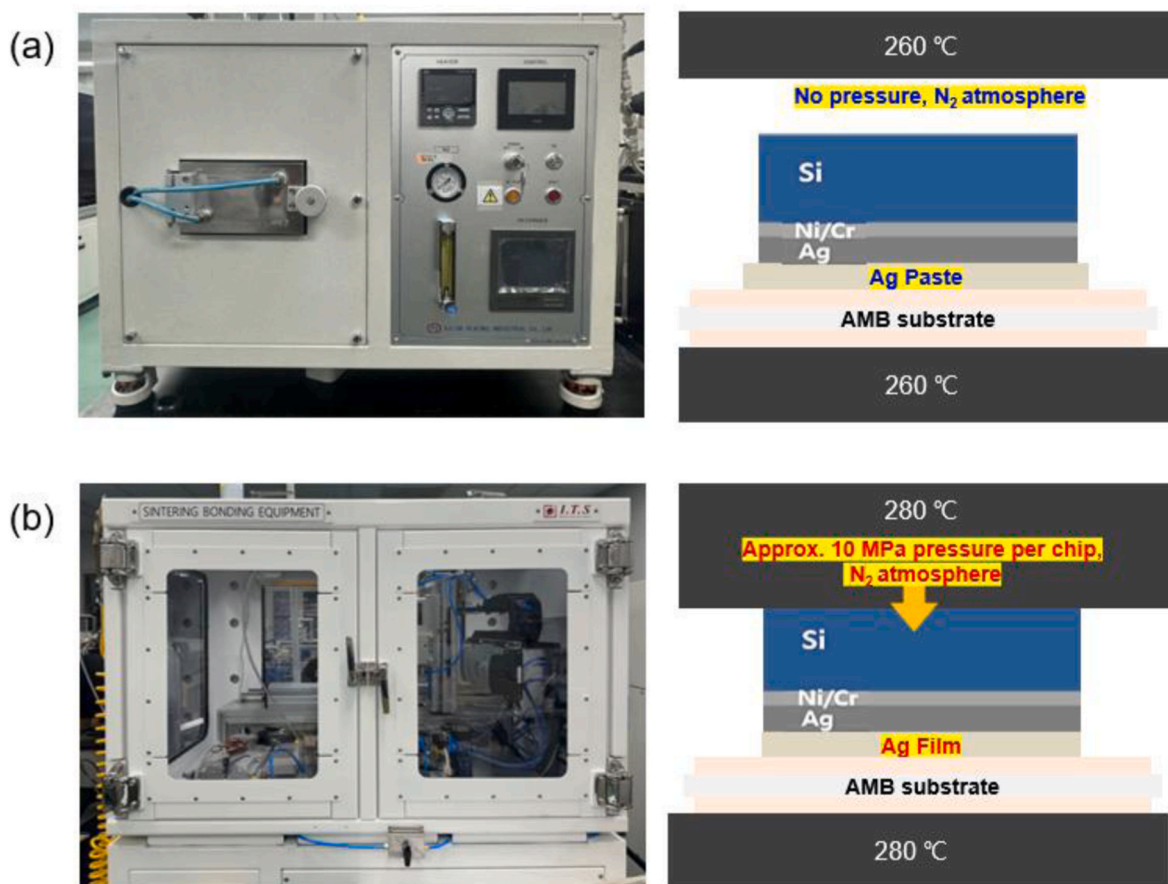


Fig. 5. Equipment and schematics for the sintering process using (a) Ag paste and (b) Ag film.

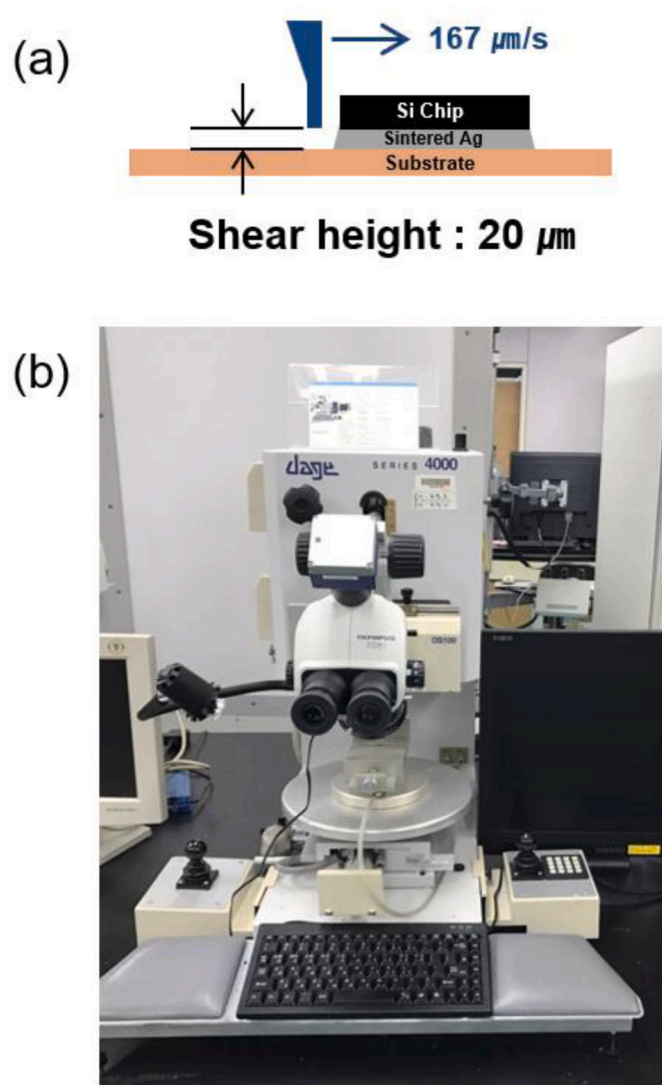


Fig. 6. (a) Schematic of the shear test and (b) Shear test equipment (Dage Series 4000, Dage Precision Industries).

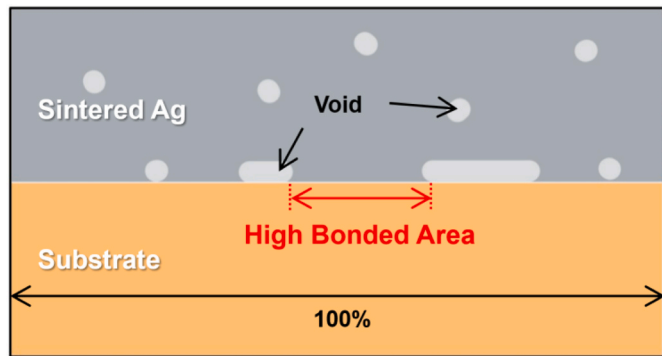


Fig. 7. Measurement method of high bond ratio (HBR).

(1.77 μm). The shear strength and internal void content (%) of the bonding interface using the Ag paste are shown in Fig. 9. For manufacturer A, the measured shear strengths on the Cu and Ag surfaces were approximately 16 and 45 MPa, respectively. By contrast, for manufacturer B, the shear strengths on the Cu and Ag substrates were 18 and 46 MPa, respectively. This generally implies higher shear strength on the

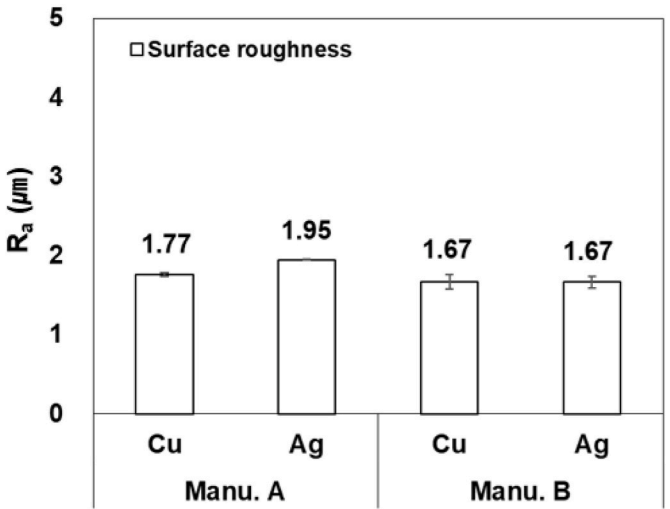


Fig. 8. Average surface roughness (R_a) values for Cu and Ag surfaces from manufacturers A and B.

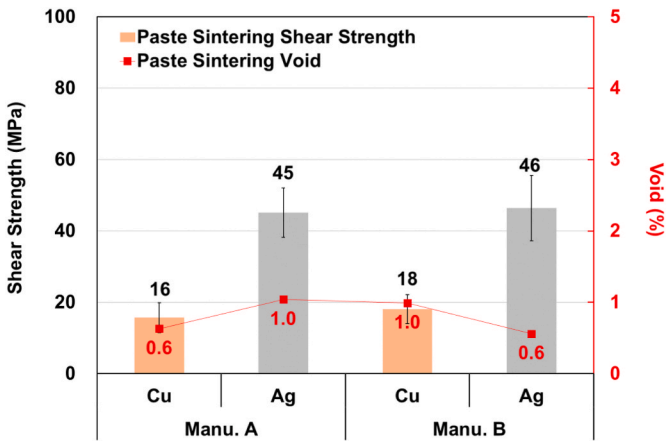


Fig. 9. Shear strength and void ratio for Ag-paste sintering.

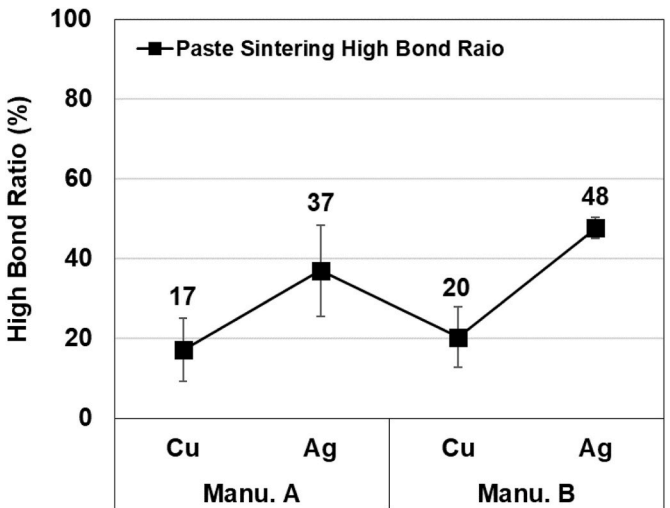


Fig. 10. High bond ratio (HBR) for Ag-paste sintering.

Ag surfaces. Typically, increased metal surface roughness is expected to enhance mechanical anchoring effects; however, despite the lower

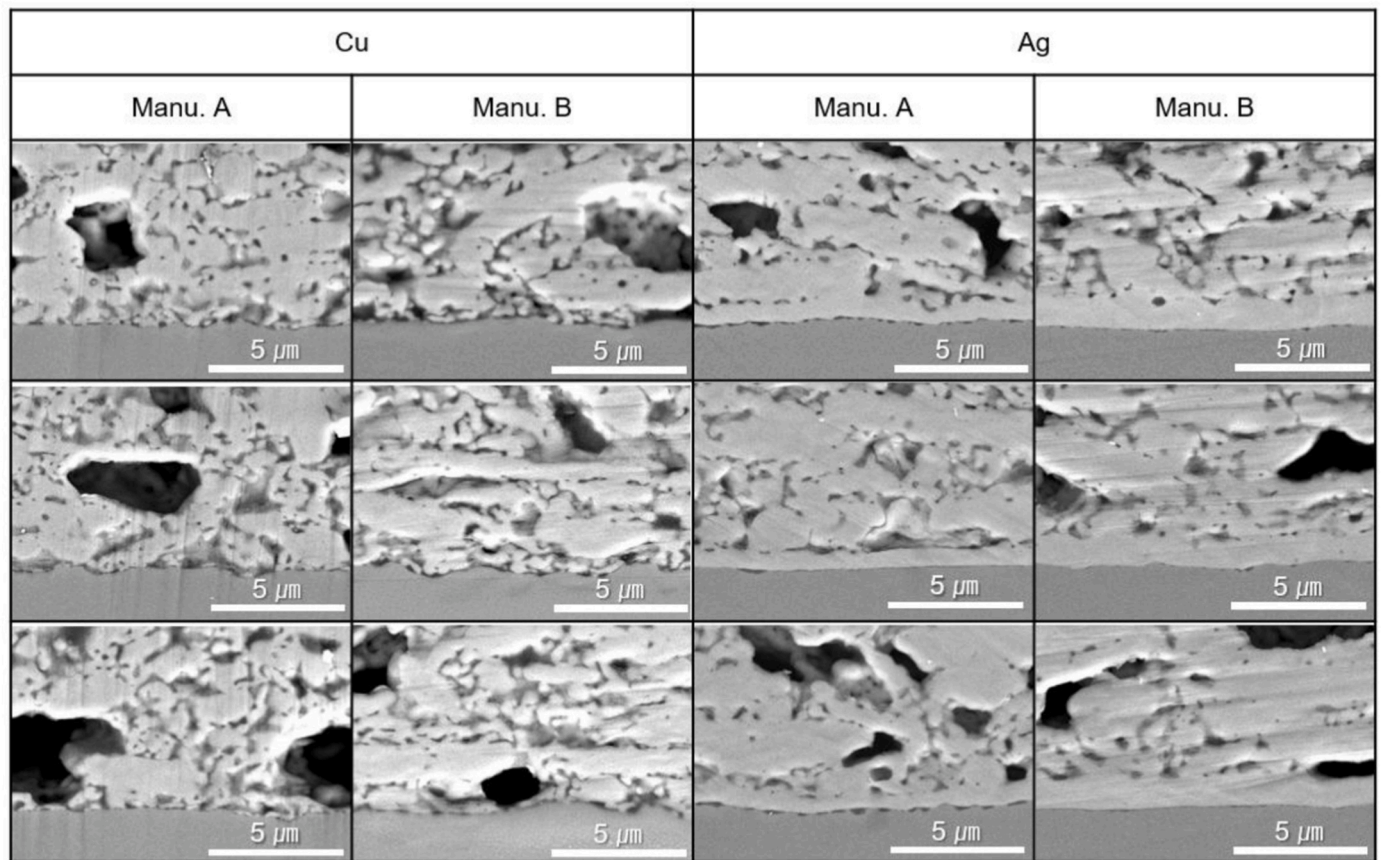


Fig. 11. Scanning electron microscopy (SEM) images of bond interfaces after Ag-paste sintering.

roughness of manufacturer B's Cu and Ag substrates compared with those of A's, they exhibited higher shear strengths [11,12]. This indicates that the effect of roughness on the bonding strength is not straightforward and is influenced by multiple factors, such as microstructure evolution, oxidation, and pressure conditions during the sintering process [13,14].

The void (%) analysis revealed that the void contents on the Cu and Ag-substrate surfaces from manufacturer A were approximately 0.6% and 1.0%, respectively, with ranges similar to those observed for manufacturer B (Fig. 9). The SEM analysis revealed localized non-uniform bonding on the Ag surface of manufacturer A. X-ray imaging showed that the voids formed during Ag-paste bonding were relatively large and clearly detectable. These voids likely formed either from residual gaps between the particles due to incomplete densification during sintering or from rapid expansion and contraction during the process [15].

Figs. 10 and 11 illustrate the HBR of Ag paste sintering and SEM microstructure of the bonded interfaces, respectively. Fig. 10 shows that manufacturer B's Ag substrate exhibited the highest HBR, suggesting a superior bonding area compared to manufacturer A. The corresponding SEM images in Fig. 11 show more uniform bonding and fewer voids at the Ag-substrate interfaces from manufacturer B, supporting the quantitative HBR results. This indicates that the effective interfacial bonding strongly depends on the controlled surface conditions rather than solely on the roughness differences. This demonstrates that higher surface roughness does not necessarily lead to better bonding. Oxide layers or impurities may accumulate in surface grooves, reducing the actual metal-to-metal bonding areas [16,17]. Moreover, manufacturer B possibly optimized various factors, such as sintering temperature, duration, internal microstructure, and oxidation suppression, ultimately achieving higher shear strength. This finding suggests that in

high-reliability electronic component bonding, controlling the entire sintering process systematically is as important as managing surface properties.

Fig. 12(a) presents SEM images of fracture surfaces from shear-tested samples where Ag paste was applied on both Cu and Ag substrates. The "Max" condition corresponds to higher shear strength, whereas the "Min" condition corresponds to a lower shear strength. Under the "Max" condition, the Ag substrate exhibits a more pronounced mixed (or combined) fracture mode that involves both the substrate–joint interface and the interior of the joint, reflecting a stronger interface and higher overall bond strength compared to the Cu substrate. By contrast, although cohesive cracking is still observed within the paste on the Cu substrate, the crack propagation appears relatively simpler. Under the "Min" condition, both substrates predominantly failed at the substrate–joint interface, indicating weaker bonding. These findings suggest that enhanced interfacial strength and reduced void formation lead to more complex crack paths and improved shear performance, in agreement with previous void and HBR analyses [18]. Furthermore, the higher shear strengths observed on the Ag substrate underscored the critical role of the substrate surface condition and microstructural control in achieving robust bonding.

As shown in Fig. 13, the shear strengths of the samples sintered using the Ag film were measured as 41 and 55 MPa on the Cu and Ag substrates, respectively, for manufacturer A, and 41 and 67 MPa on the Cu and Ag substrates, respectively, for manufacturer B. The corresponding HBR values after sintering are presented in Fig. 14, indicating bonding ratios of 31% (Cu substrate) and 37% (Ag substrate) for manufacturer A and 36% (Cu substrate) and 62% (Ag substrate) for manufacturer B. The notably higher HBR observed for manufacturer B's Ag substrate can be attributed to its lower surface roughness, which significantly enhanced the interfacial uniformity and effectively suppressed interfacial

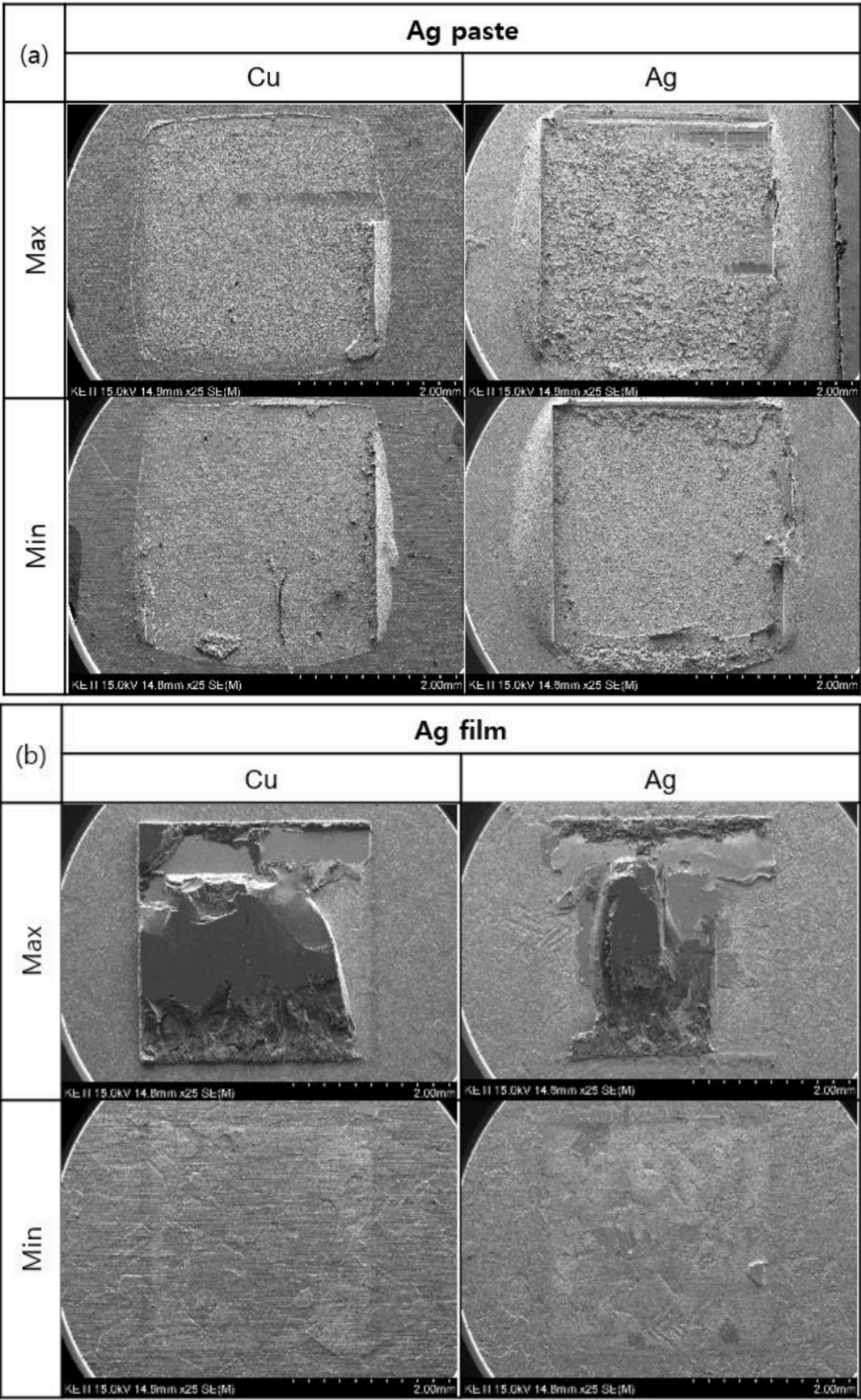


Fig. 12. Scanning electron microscopy (SEM) analysis of fracture surfaces from shear-tested samples: (a) Ag paste, (b) Ag film.

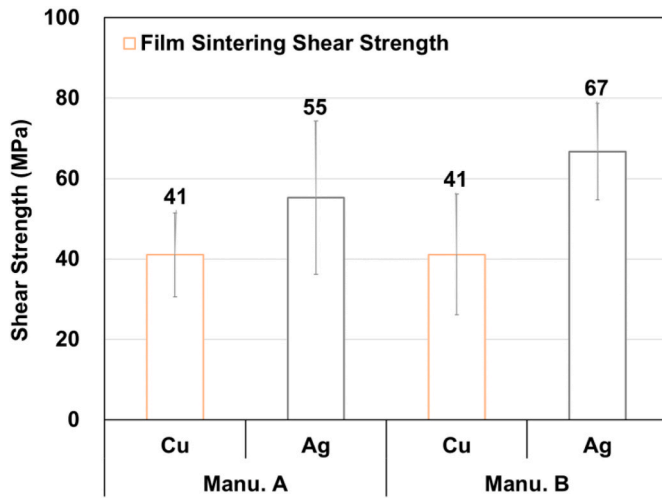


Fig. 13. Shear strength for Ag-film sintering.

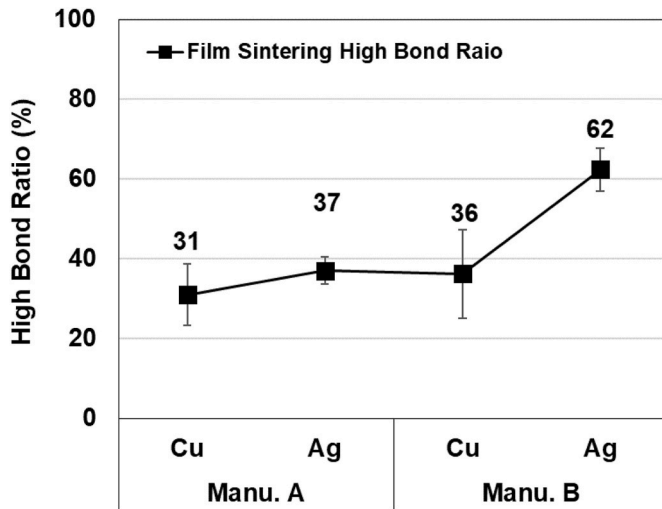


Fig. 14. High bond ratio (HBR) for Ag-film sintering.

oxidation under pressure-assisted film sintering. Furthermore, the Ag film enabled enhanced interfacial metal interdiffusion under pressure, promoting efficient bonding area formation. These findings demonstrate that lower surface-roughness conditions are more favorable for achieving superior bonding quality in pressure-assisted film sintering [6].

A detailed examination of the interface morphology in the SEM images (Fig. 15) shows that, unlike paste bonding, large voids were nearly absent in the film-bonding layer, although micro-scale voids are partially observed. Additionally, owing to the small sizes of these voids, quantitative evaluation using X-ray analysis was challenging, posing limitations to calculating the actual void fraction. Therefore, the void ratios measured by X-ray analysis only represent voids larger than the X-ray detection limit ($\sim 30 \mu\text{m}$). Thus, the assertion that Ag-film bonding “suppresses voids” primarily refers to the reduction of large, X-ray-detectable voids, while micro-voids observed by SEM may still remain. This constraint should be considered when comparing void-related metrics between paste and film bonding. The X-ray analysis results (Fig. 16) also indicate that voids were difficult to observe in the film-bonding layer, with an overall uniform contrast. This is likely because film-based materials are formed with a predetermined thin-film thickness during the initial deposition, and even after sintering, the probability of large void formation owing to rapid shrinkage or binder

evaporation is relatively low [10]. However, because micro-voids were observed in the SEM images, further optimization of the sintering temperature and atmospheric conditions is necessary to enhance interface adhesion and control micro-void formation.

Fig. 12(b) presents the SEM images of the fracture surfaces observed in the shear-tested specimens bonded with the Ag film. Compared to the Ag-paste bonding results shown in Fig. 12(a), specimens showing higher shear strength tend to exhibit fracture regions extending toward the chip interface, resulting in multiple fracture sites. Conversely, those with lower shear strengths failed predominantly at the substrate interface. This suggests that, under high shear strength conditions, the interfacial bond is sufficiently robust to allow fractures to propagate to the chip–film interface, which is consistent with the lower void content and more uniform contact observed in Ag-film bonding compared with the Ag paste approach illustrated in Fig. 11. Therefore, the film-based method demonstrates a superior bonding strength, where fracture can occur on the chip side instead of remaining confined to the substrate side at higher shear loads, highlighting a distinct advantage for high-reliability electronic component design. Notably, the reliability implications discussed in this study are based on as-sintered interfacial integrity and shear-fracture behavior under the present test conditions. External durability validations under service-like conditions (e.g., thermal cycling, high-temperature aging, or power cycling) were not performed and can influence the relative performance trends. Thus, the reliability-related conclusions should be interpreted within the scope of the current mechanical tests and microstructural observations. Fig. 17 schematically illustrates the main crack locations observed on the fracture surfaces after shear testing. In paste sintering bonding (a), the main crack primarily occurs at the interface between the substrate and sintered Ag layer, whereas in film sintering bonding (b), cracks predominantly occur at the chip–Ag layer interface. This clearly demonstrates that film bonding provides superior interfacial adhesion, causing the fracture location to shift toward the chip under high-load conditions.

Overall, Ag-film bonding significantly suppressed void formation compared with paste bonding. Even with some micro-voids present, it achieved a uniform interface structure and high shear strength. Particularly, when combined with Ag substrates, favorable surface properties and enhanced metal interdiffusion contributed to a higher bond strength. Further research and process optimization will help advance reliable bonding technologies.

3.2. Microstructural characteristics and their impact on bond reliability

Fig. 18 shows the average grain sizes of the Ag films sintered on Cu and Ag substrates at the bonding interface. The EBSD analysis established that the average grain size of the Ag film bonded to the Cu substrates was $0.481 \mu\text{m}$, while that of the Ag film bonded to the Ag substrates was $0.426 \mu\text{m}$. The larger grain size on the Cu substrates suggests that Cu promoted grain growth in Ag films during sintering. In the early stage of sintering, localized metal reactions may accelerate recrystallization, leading to a dominant grain growth [19–21].

In contrast, the smaller grain size observed on the Ag substrates indicates that the sintered layer formed more uniformly due to even interface reactions within the same metal system [21,22]. Overall, the grain growth appeared uniform, with slightly smaller average grain sizes in the Ag-substrate samples than in the Cu substrates, thus maintaining a fine-grained structure while achieving densification. Such a fine-grained structure can benefit reliability by creating complex crack-propagation paths [23].

The EBSD analysis was conducted to examine the distribution of grain misorientation at the bonding interface after Ag-film bonding. Fig. 19 illustrates the misorientation distribution at the Cu- and Ag-bonding interfaces after sintering with the Ag film. The box plot in Fig. 20 quantitatively compares the distribution of these misorientation values. The EBSD results in Fig. 20 show that the mean misorientation angles for Ag films bonded to Cu and Ag substrates were approximately

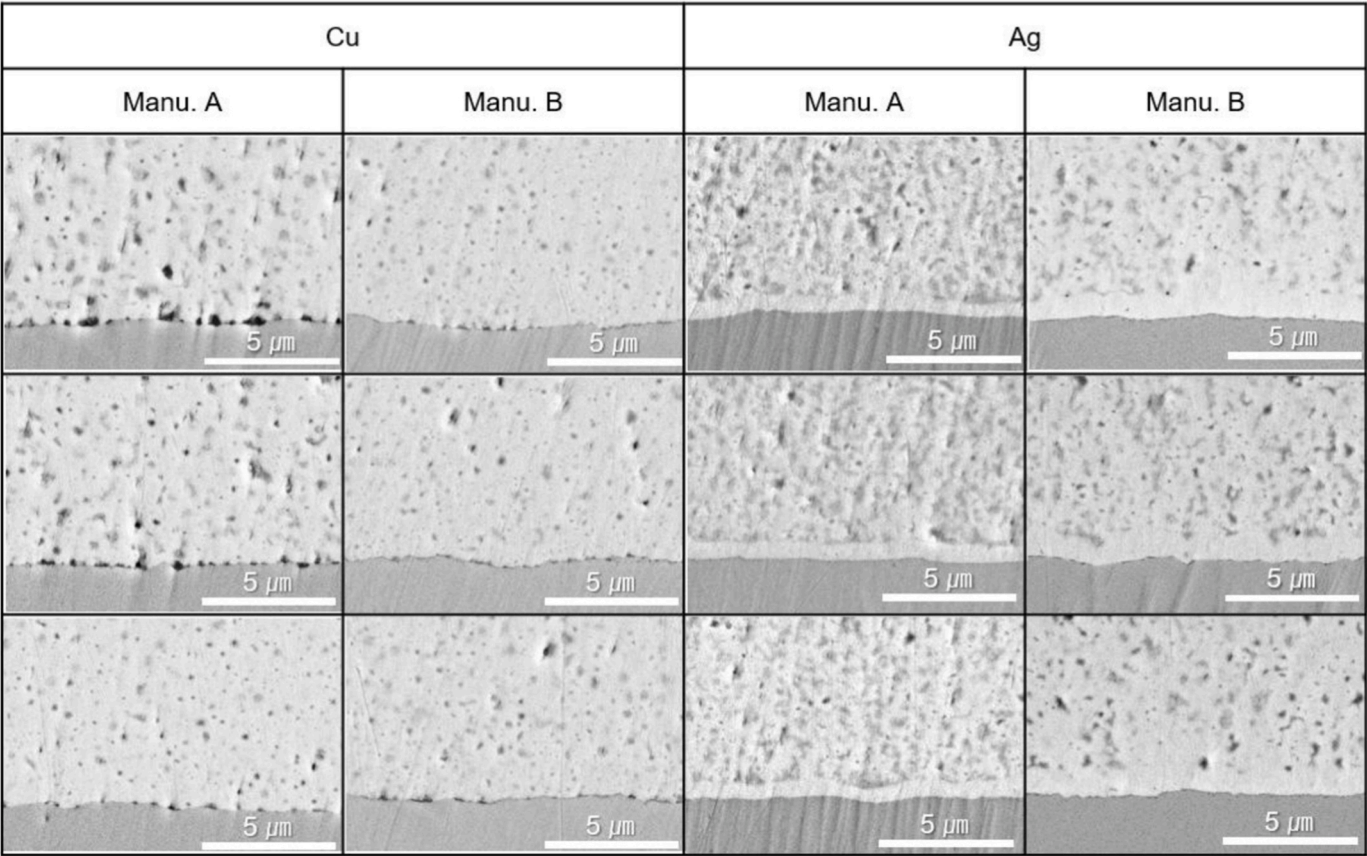


Fig. 15. Scanning electron microscopy (SEM) images of bond interfaces after Ag-film sintering.

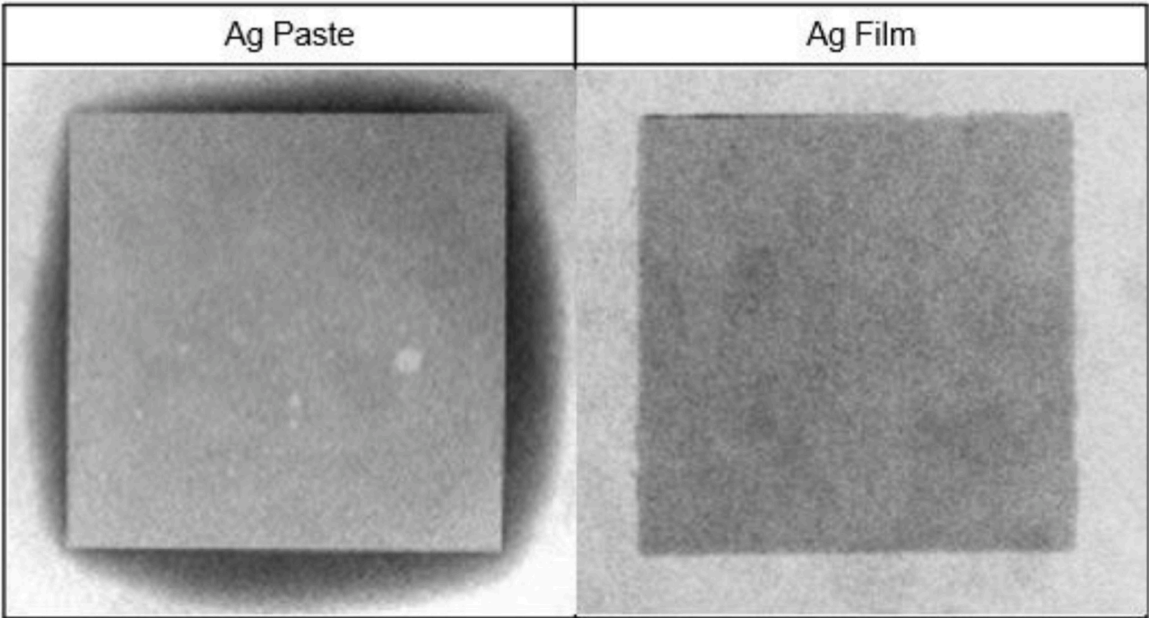


Fig. 16. X-ray analysis results after sintering.

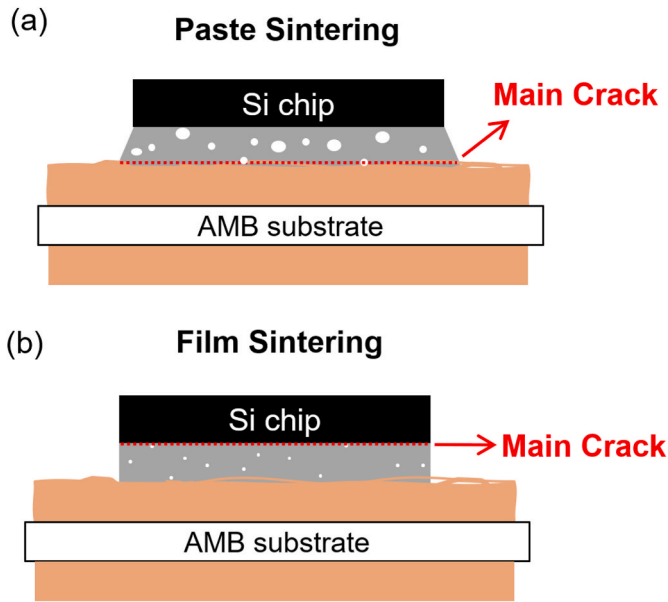


Fig. 17. Schematic showing the main crack locations in fracture surfaces after shear tests: (a) Paste sintering, (b) Film sintering.

36° and 37°, respectively, with no significant difference. However, a closer look at the distribution reveals that the misorientation angles on Cu substrates ranged from 20° to 57°, whereas those on the Ag substrates were more concentrated in the range of 30° to 45°. This suggests that different metal systems (Cu and Ag) caused diffusion and recrystallization to proceed differently in the localized regions, leading to a wider variation in grain-boundary angles [24]. Although this broad distribution may aid in local stress dispersion at the bonding interface, it could also reduce interface uniformity and potentially degrade reliability [25]. Some studies have reported interface degradation in Cu–Ag bonding interfaces after thermal cycling tests and proposed methods, such as pre-activating or alloying Cu surfaces to control diffusion behavior [26–28]. Thus, achieving sufficiently high bond strength in Cu-substrate samples may require optimizing the sintering conditions (temperature and time) or implementing post-treatment processes to enhance and control the Cu–Ag interfacial reactions [29–31]. Because the Ag–Ag interface involves bonding between identical metals, diffusion and grain rearrangement proceed more actively and uniformly during the sintering process, resulting in uniform grain growth and a narrower distribution [32]. Under these conditions, the interface structure exhibits enhanced consistency, which is advantageous for long-term reliability. This consistency simultaneously contributes to improvements in both the strength and thermal stability of the sintered layer, making it one of the key reasons why Ag-film sintering is preferred in the high-power electronics manufacturing sector. In conclusion, an optimal balance of process conditions, such as sintering temperature, time, and atmosphere, along with control of interfacial reactions (diffusion and alloying) is necessary to leverage the advantages of each substrate [33].

The influence of surface roughness from different manufacturers on shear strength varied significantly depending on the sintering methods applied [34]. Under pressure, less sintering (Ag paste) and substrates with higher surface roughness exhibited enhanced shear strength owing to the increased effective contact area, promoting improved interfacial

adhesion [35]. Conversely, under pressure-assisted sintering (Ag film), substrates with lower surface roughness demonstrated superior shear strength owing to improved interfacial uniformity and densification of the bonding interface. The EBSD analysis further validated these results by revealing distinct grain size distributions and misorientation characteristics, indicating that lower surface roughness significantly enhanced interfacial structural integrity under pressure-assisted conditions.

Our observations on the roughness sensitivity of Ag-paste bonding are consistent with previous reports showing that pressureless Ag paste strongly depends on surface topography and oxidation conditions [5,7]. In addition, the suppressed void formation and enhanced interface densification observed in Ag-film bonding align well with earlier studies on pressure-assisted nanosilver films [6,10]. The fracture-location shift toward the chip side at higher interfacial integrity also agrees with prior analyses on microporous Ag joints under mechanical loading [18]. Compared with these studies, our work uniquely integrates supplier-dependent AMB surface conditions with EBSD characterization to provide a more comprehensive interpretation of interfacial uniformity and bonding performance. Compared with previous studies that often investigate either a single bonding form or a single substrate condition, the present work unifies supplier-dependent AMB surfaces (A vs. B), Cu/Ag finishes, and two Ag sintering forms (pressureless paste vs. pressure-assisted film) within one experimental framework. By combining X-ray- and SEM-based void assessment (considering the X-ray detection limit), HBR quantification, shear-fracture location analysis, and EBSD-based crystallographic characterization, we provide an interface-informed explanation linking surface states to void evolution and joint reliability.

4. Conclusions

This study systematically analyzed the effects of surface treatments on sintered bonding characteristics of Ag paste and Ag film using Cu and Ag AMB substrates from two manufacturers (A and B). Key findings are summarized as follows.

- The Ag surface from manufacturer A exhibited superior bonding strength compared with its Cu counterpart, despite its higher roughness. The EBSD analysis showed a narrower misorientation angle distribution at the Ag/Ag interface, indicating improved microstructural uniformity and bonding quality.
- A key contribution of this study is the systematic cross-comparison of Ag paste (pressureless) and Ag film (pressure-assisted) across supplier-dependent AMB substrates (A/B) and Cu/Ag finishes, integrating X-ray and SEM void analysis, HBR, shear-fracture location, and EBSD-based microstructural characterization to interpret interfacial integrity.
- Cu substrates promoted grain growth in Ag films, leading to larger average grain sizes, although local non-uniformities were observed. In contrast, Ag substrates exhibited smaller and more uniform grain sizes and formed a denser interface, thereby ensuring a stable and high bonding quality.
- In Ag-film bonding, the Cu substrates exhibited a broader range of grain orientations, offering flexibility but potentially compromising the uniformity and mechanical stability. Moreover, Ag substrates achieved superior bonding quality due to their higher uniformity and structural stability.

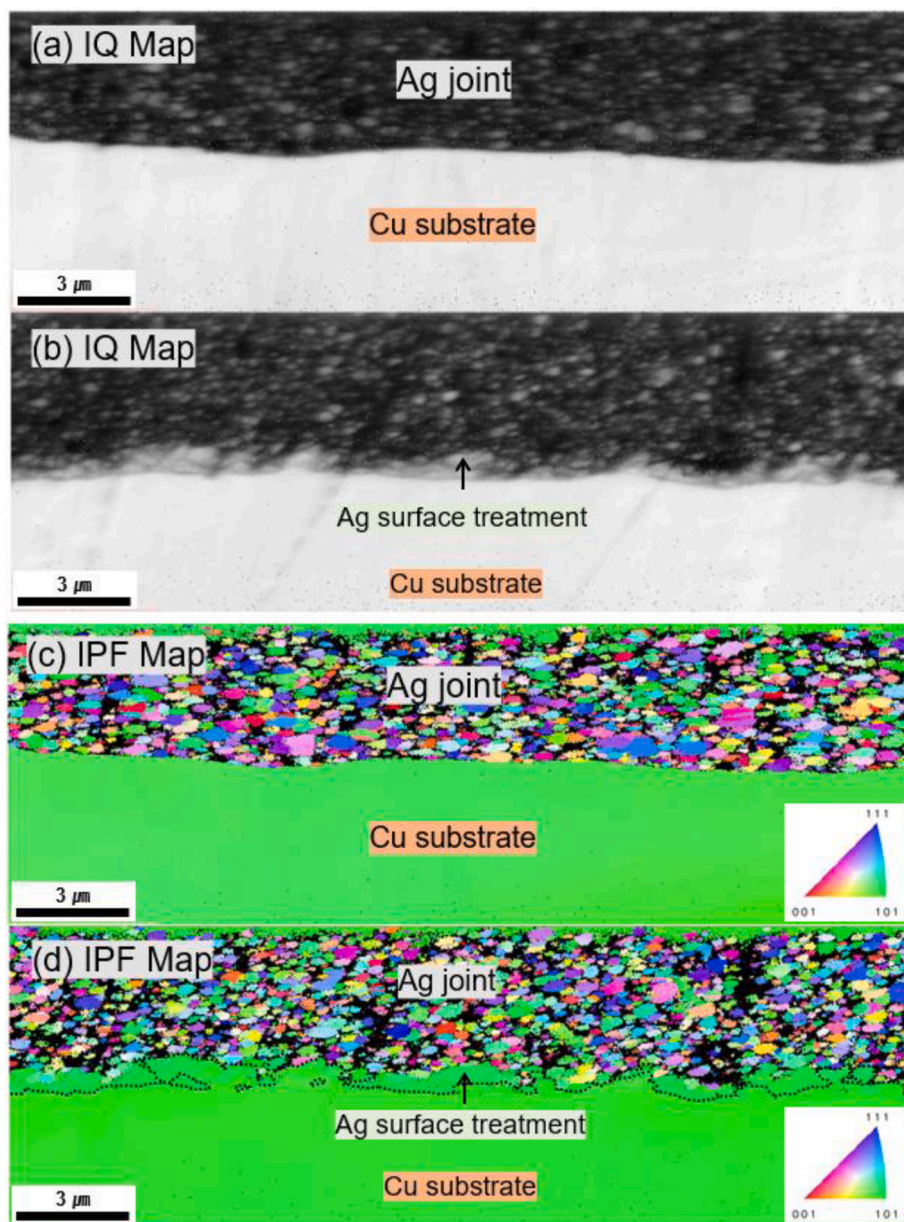


Fig. 18. Image quality (IQ) and inverse pole figure (IPF) maps of the bonding interface with Ag film after sintering: (a), (b) Cu and Ag substrate IQ maps, (c), (d) Cu and Ag substrate IPF maps.

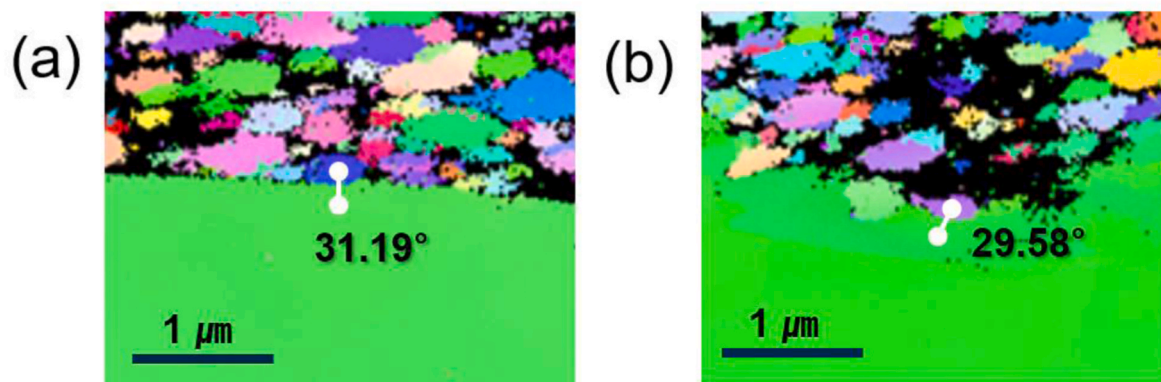


Fig. 19. Misorientation analysis at the bonding interface after sintering with Ag film: (a) Cu, (b) Ag.

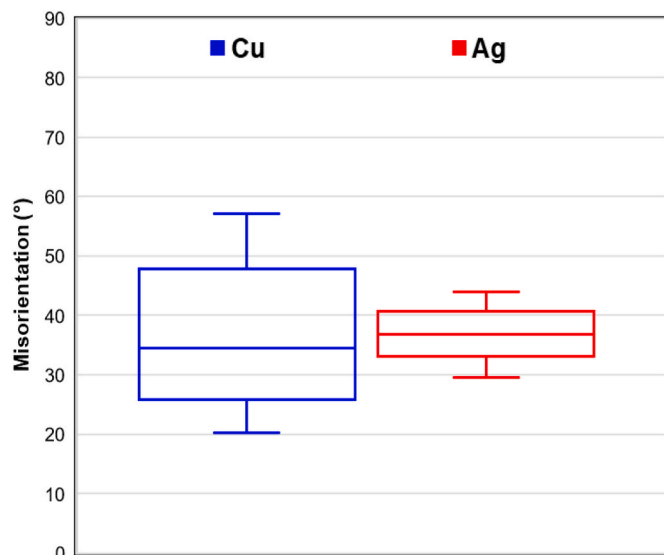


Fig. 20. Misorientation angles at the bonding interface after sintering with Ag film.

This study demonstrated that despite the higher roughness of the Ag surface from manufacturer A compared with that of Cu, the former achieved stronger bonding, proving that bonding quality is influenced not only by surface roughness but also by inherent material properties and sintering conditions. Additionally, film bonding, through pressure sintering, minimized the impact of surface roughness, enabling uniform and high-quality bonding; however, the issue of chip fractures needs to be addressed. These constraints directly qualify the interpretation and significance of the void-related comparisons and reliability discussions presented in this study. Therefore, the conclusions should be interpreted within the current measurement limits (including the X-ray detection threshold) and as-sintered test conditions. Future research should focus on optimizing Ag-film transfer and sintering processes and evaluating their applicability to various chip materials.

CRediT authorship contribution statement

Jeeyeon Park: Writing – original draft. **Jeong-Won Yoon:** Writing – review & editing, Validation. **Semin Park:** Writing – review & editing, Conceptualization. **Jihyung Lee:** Writing – review & editing, Conceptualization. **Hyojin Park:** Writing – review & editing, Data curation. **Chulmin Oh:** Writing – review & editing, Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rsufi.2026.100741>.

Data availability

Data will be made available on request.

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